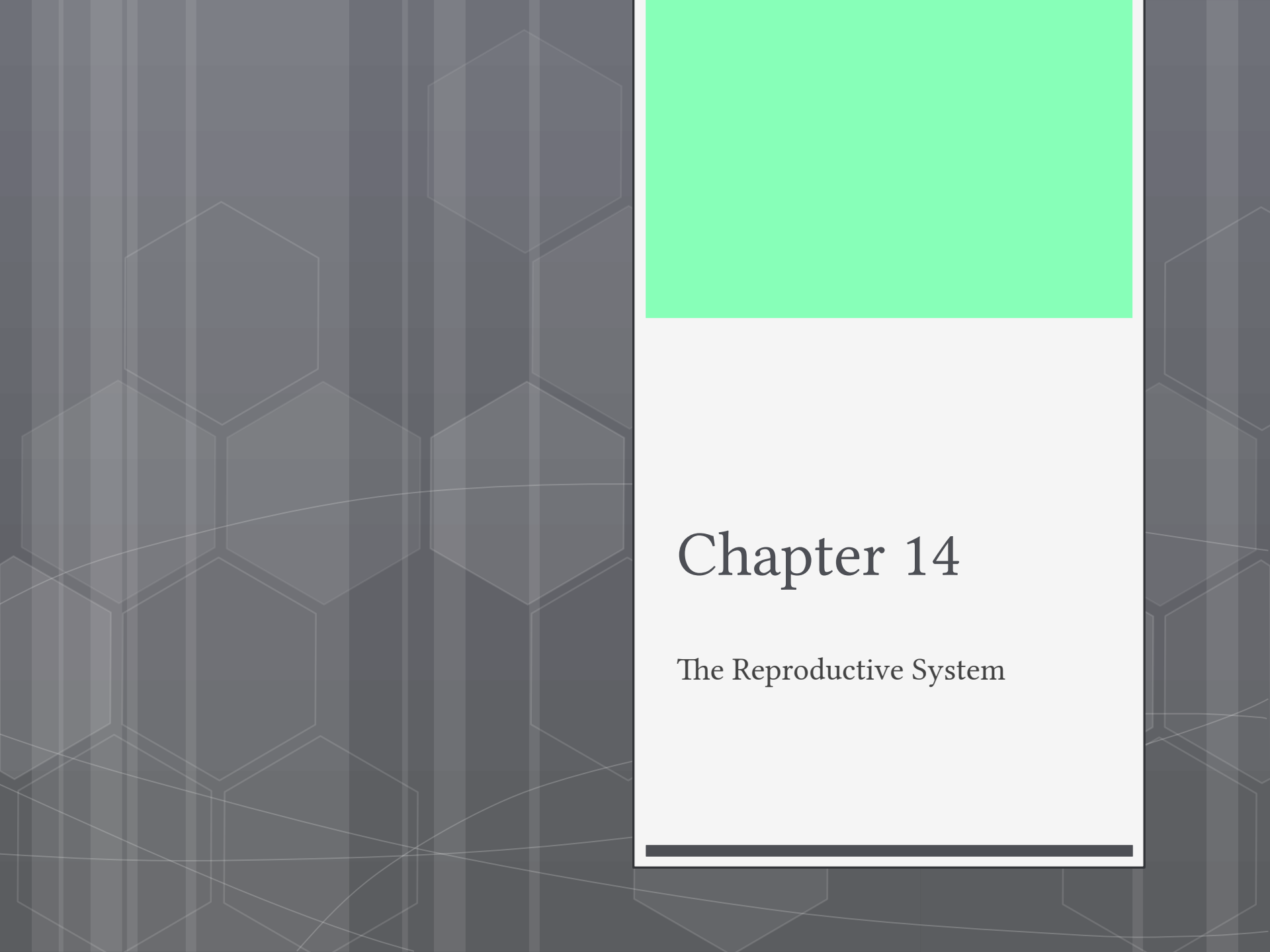


BIOLOGY 30

UNIT B:
**Reproduction &
Development**



Chapter 14

The Reproductive System

Chapter 14 The Reproductive System

- The male reproductive system (key structures, spermatogenesis, secondary sex characteristics)
- Male sex hormones (roles of testosterone, LH & FSH)
- The female reproductive system (key structures, oogenesis, secondary sex characteristics)
- Sexually transmitted infections
- Hormonal regulation of gametogenesis

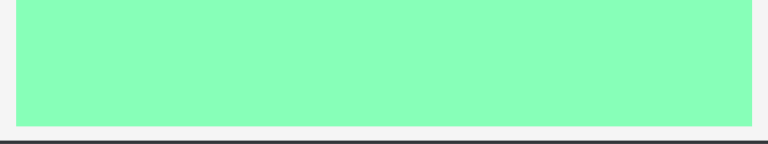
Textbook pages 476-505

Male and female reproductive systems are characterized by distinct...

- **Gonads** (gamete-producing organs → testes vs. ovaries)
- **Sex hormones** (testosterone vs. estrogen)
- **Gametes** (sex cells → sperm vs. eggs)
- **Primary sex characteristics**
 - These are the **reproductive organs/structures**
 - Typically **present at birth** but further develop during puberty
- **Secondary sex characteristics**
 - features not directly related to reproduction
 - Develop over time, especially after puberty, in response to increased sex-hormone levels
 - e.g. facial hair, voice changes, breast development, etc.

Primary vs. Secondary Sex Characteristics

Sex	Primary sex characteristics	Selected secondary sex characteristics (with wide variations among individuals and ethnic groups)
male	• gonads (testes) • penis • scrotum • seminal vesicles • epididymis • prostate gland • ductus deferens • Cowper's gland	• facial hair • body hair • deeper voice, broader shoulders, narrower hips, and more obvious muscle development compared with female
female	• gonads (ovaries) • cervix • oviducts • vagina • uterus • vulva	• minimal facial hair • minimal body hair • prominent breasts (compared with male) • higher voice, more rounded shoulders, wider hips, and less obvious muscle development compared with male



Control of Reproductive System

Sex Determination

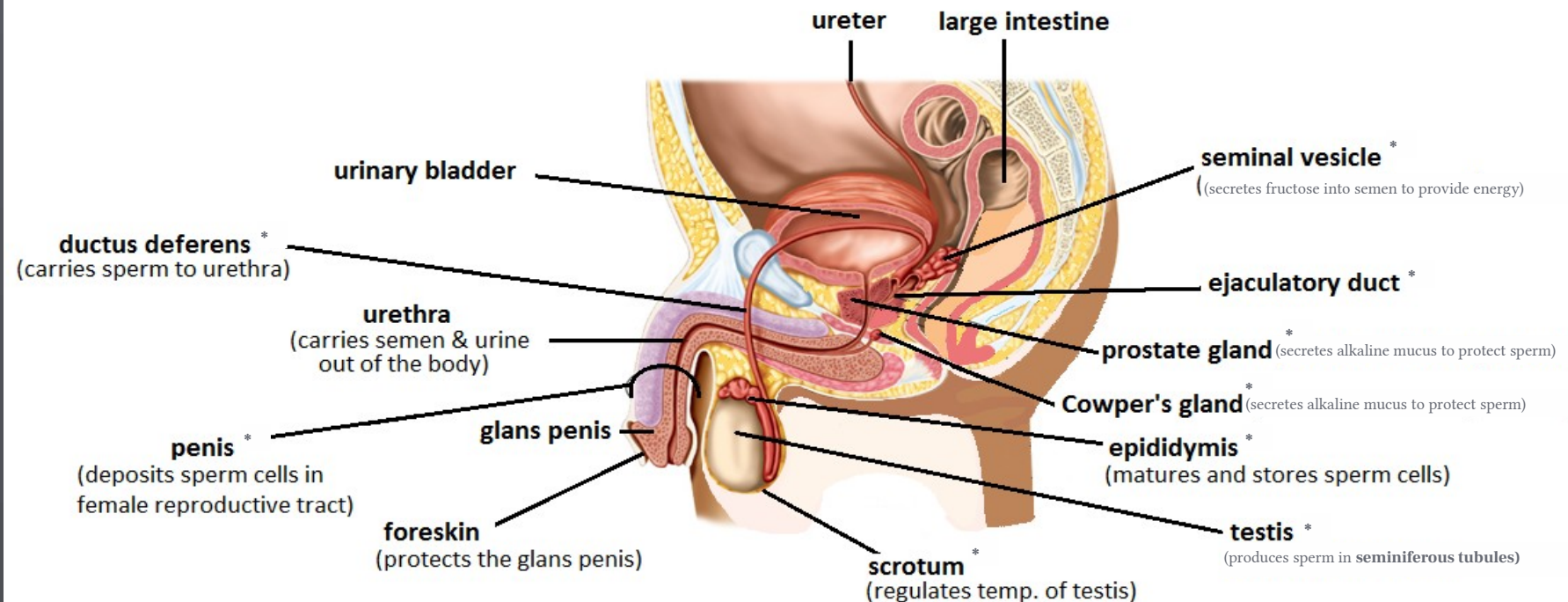
- How do embryos develop into male or female?
- There are 23 pairs of chromosomes in humans (46 total)
- The 23rd pair is the sex chromosome pair, and are different between male/female
 - Females have two 'X' chromosomes (XX)
 - Males have an X and a Y chromosome (XY)
- The presence of a Y chromosome stimulates the production of testes
 - Testes in turn secrete testosterone which leads to the development of male sex characteristics 6-7 weeks after fertilization

The Male Reproductive System



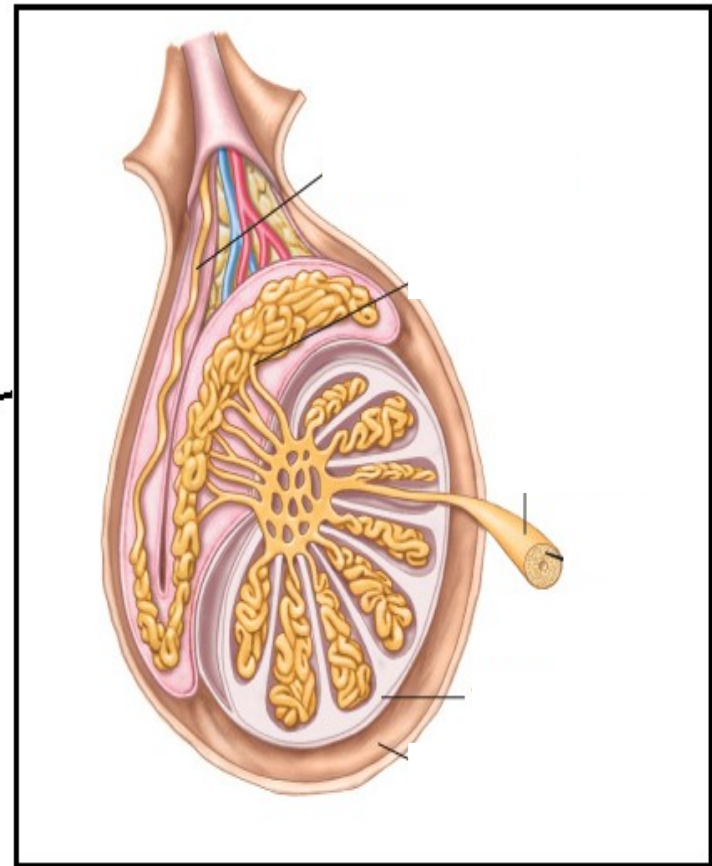
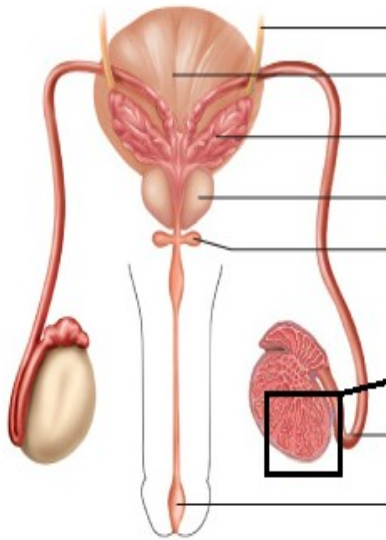
The Male Reproductive System

**you need to know all these structures and functions. A more comprehensive table with functions is on a later slide.*

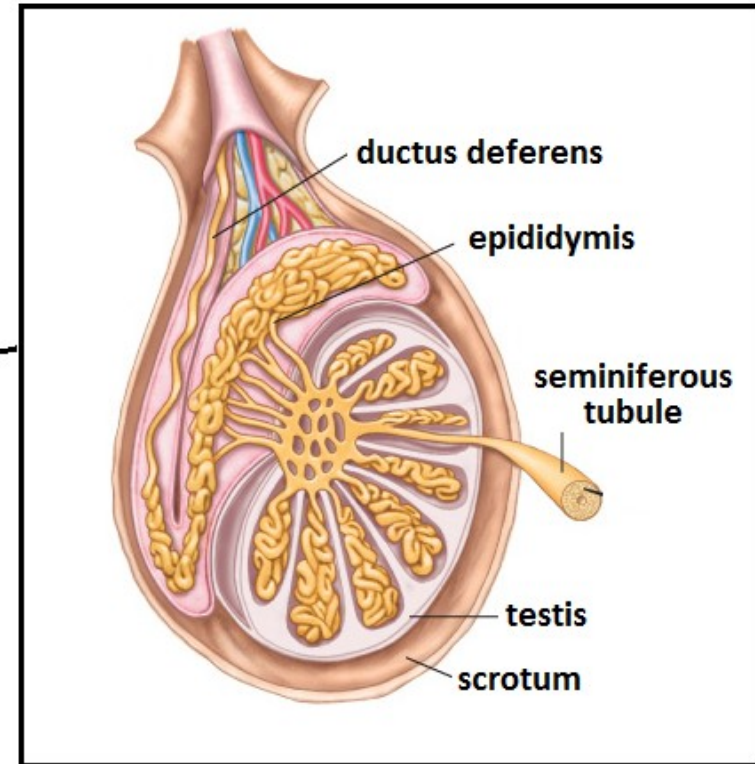
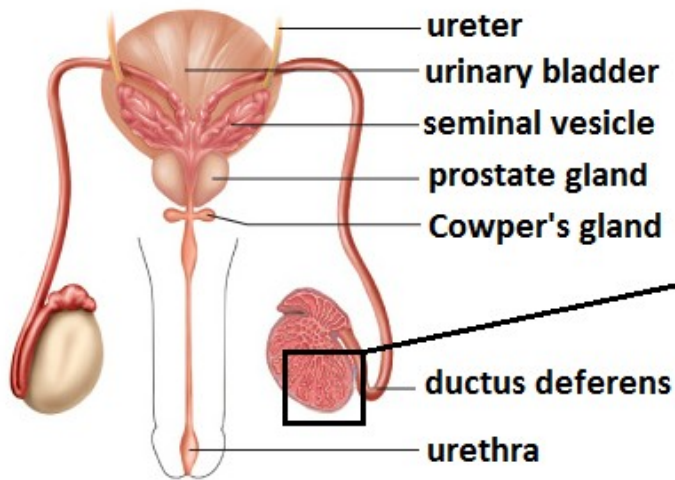


Can you match the structures on the previous page to a front view diagram?

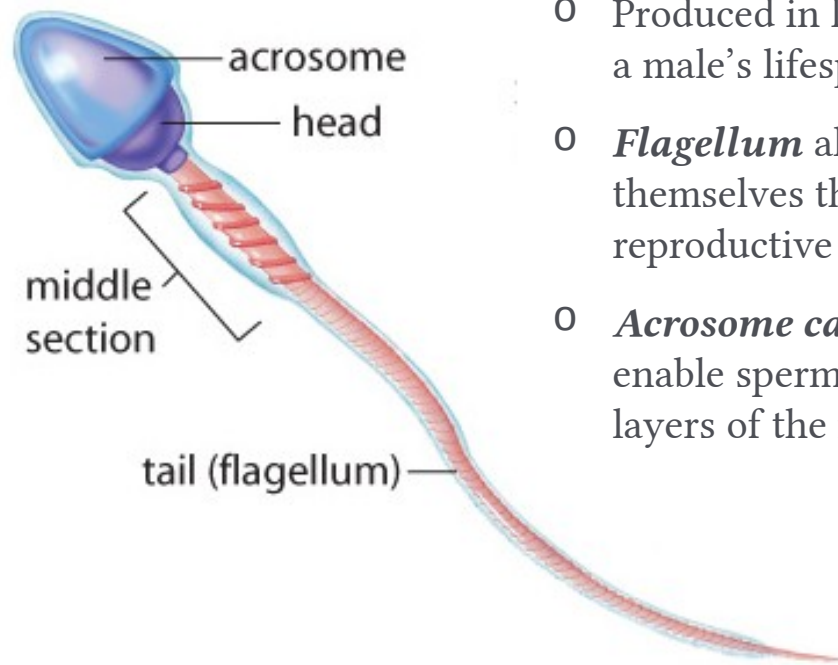
(Answers on the next slide)



Male Reproductive Structures (Front View)

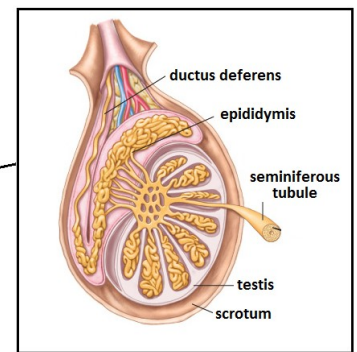
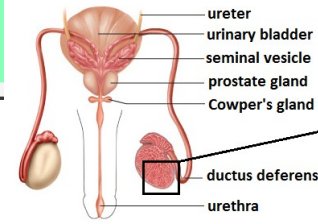


Sperm: The Male Gamete (sex cells)

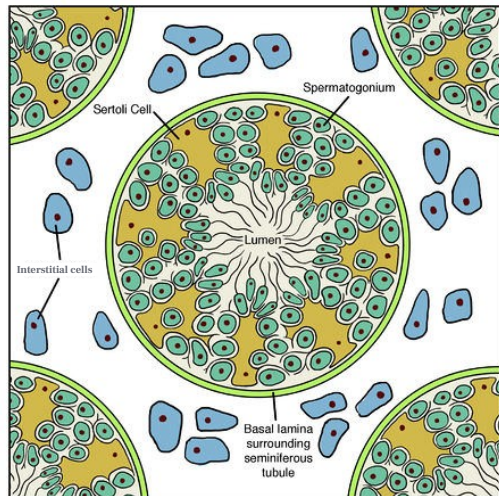


- Produced in large amounts throughout a male's lifespan starting at puberty
- **Flagellum** allows sperm to propel themselves through the female reproductive system
- **Acrosome cap** releases enzymes which enable sperm to penetrate the outer layers of the female egg

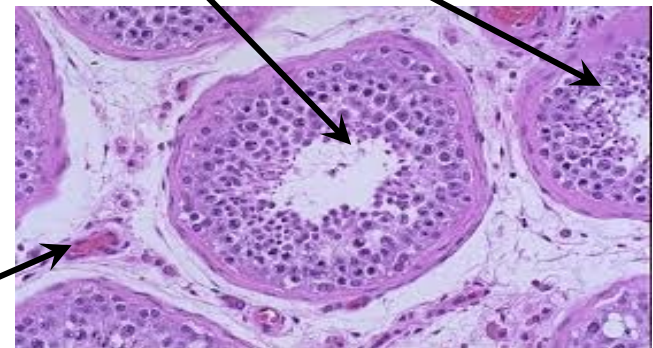
SPERMATOGENESIS (Sperm production)



- The testes are composed of long, coiled tubes called *seminiferous tubules*; this is where sperm are produced during **spermatogenesis**
- Testes also contain testosterone-secreting cells called *interstitial cells* (also known as Leydig cells), that surround the seminiferous tubules. The **testosterone** released by the interstitial cells is also indirectly involved in spermatogenesis by stimulating various functions of Sertoli cells.



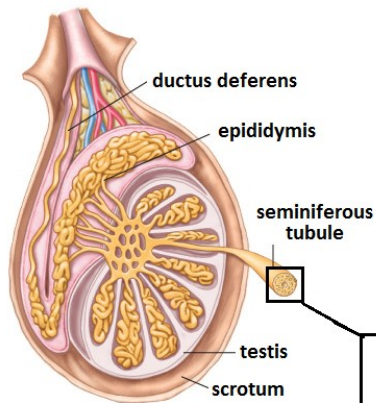
Seminiferous tubule (cross section)



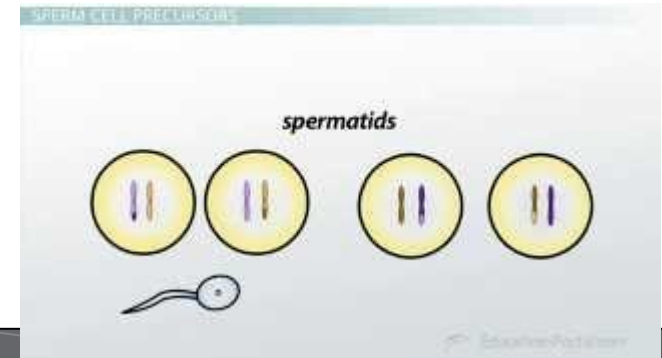
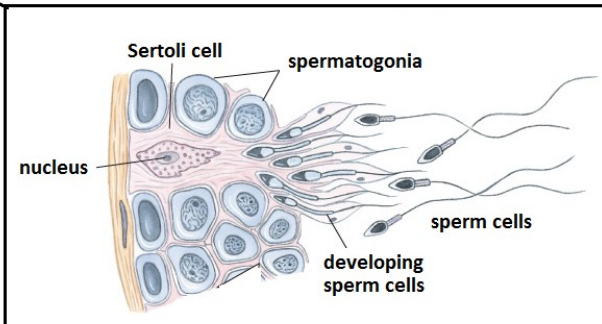
Interstitial cell

SPERMATOGENESIS

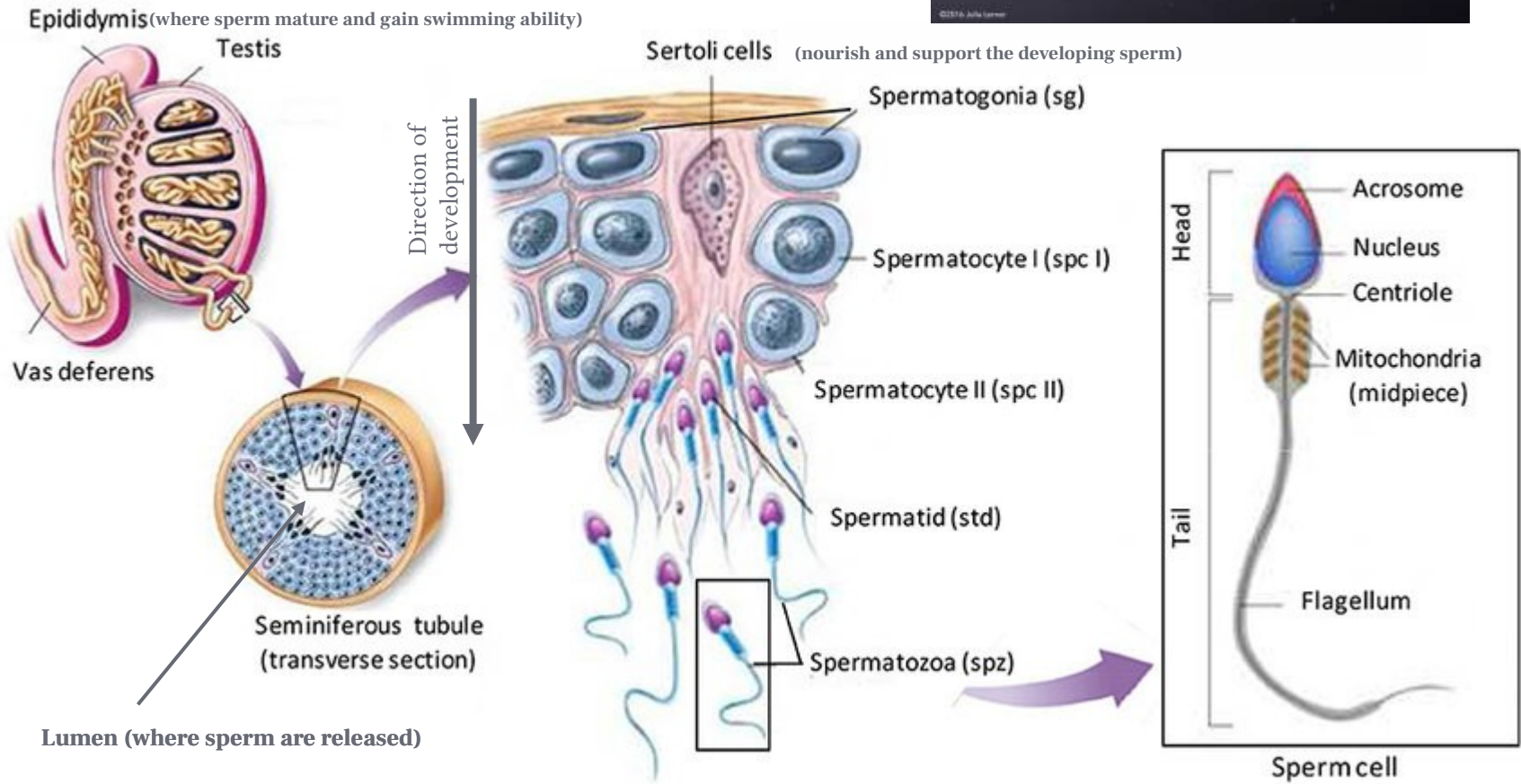
- Sperm develop from precursor cells called **spermatogonia** within seminiferous tubules
- Spermatogonia undergo two cycles of meiosis (more on this in Chapter 16) as they develop into a primary spermatocyte, then a secondary spermatocyte, and ultimately a **spermatid**.
- Spermatids then go on to differentiate into **spermatozoa (sperm)**
- These series of cell divisions and differentiation occur towards the middle of the seminiferous tubule, where the final spermatocytes are released into the **lumen**



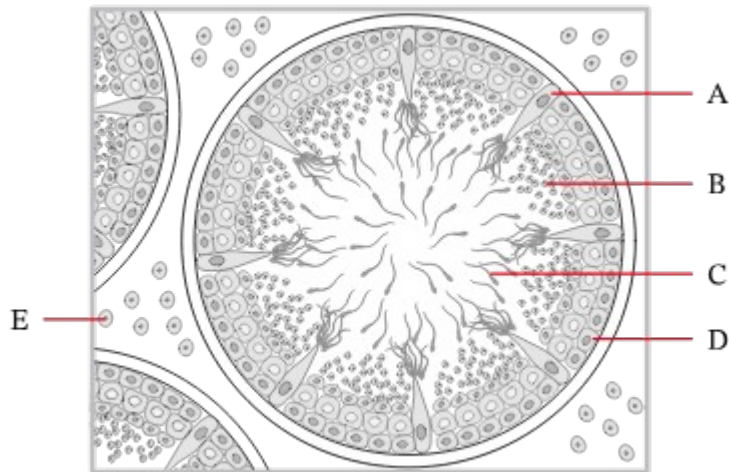
- **Sertoli cells** within the seminiferous tubules nourish and provide metabolic support for developing sperm. These can be easily recognized on diagrams as the largest cells in the tubule.
- Sperm cells mature and gain their ability to swim in the **epididymis** of the testes



SPERMATOGENESIS



Can you label the following structures?



- A. Sertoli cell
- B. Spermatid
- C. Spermatozoa (Sperm)
- D. Spermatogonium
- E. Interstitial cell

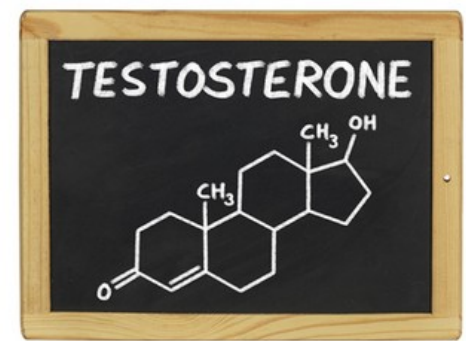
Male Sex Hormones

Male sex hormones are known as *androgens*. While **testosterone** is the primary male sex hormone, a series of other hormones are required for spermatogenesis.

- **Testosterone** is the hormone that initiates the development of male sex organs and ducts
- When a male reaches puberty, testosterone levels increase, leading to the development of secondary sex characteristics
- Testosterone also plays a central role in spermatogenesis

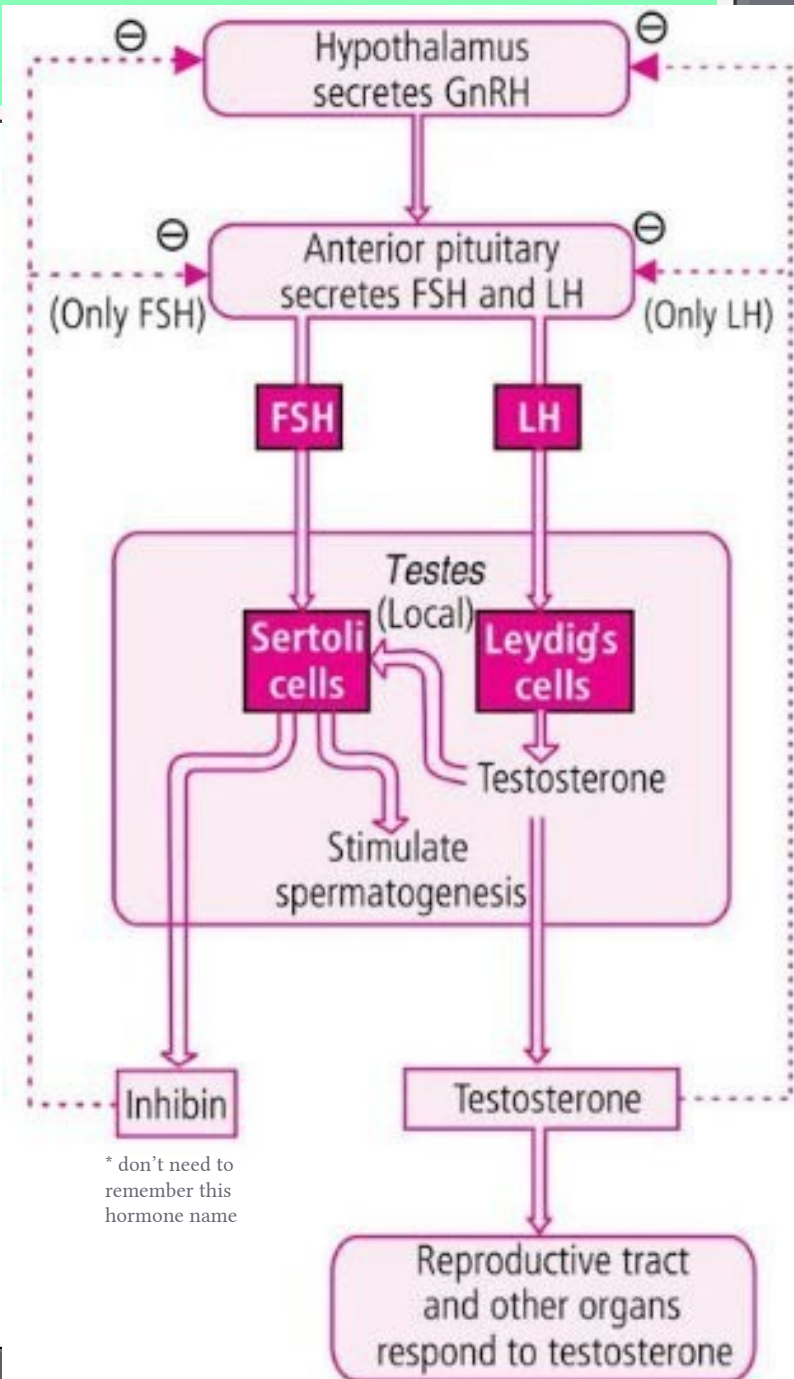
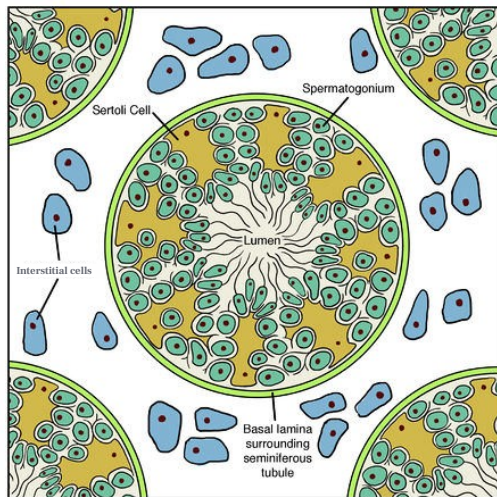
The following table represents blood testosterone levels at different stages in a man's life:

Age (years)	Blood testosterone level (ng/dL)
1 to 7.9	40
8 to 10.9	42
11 to 11.9	260
12 to 13.9	420
14 to 17.9	1000
18 to 29	1100



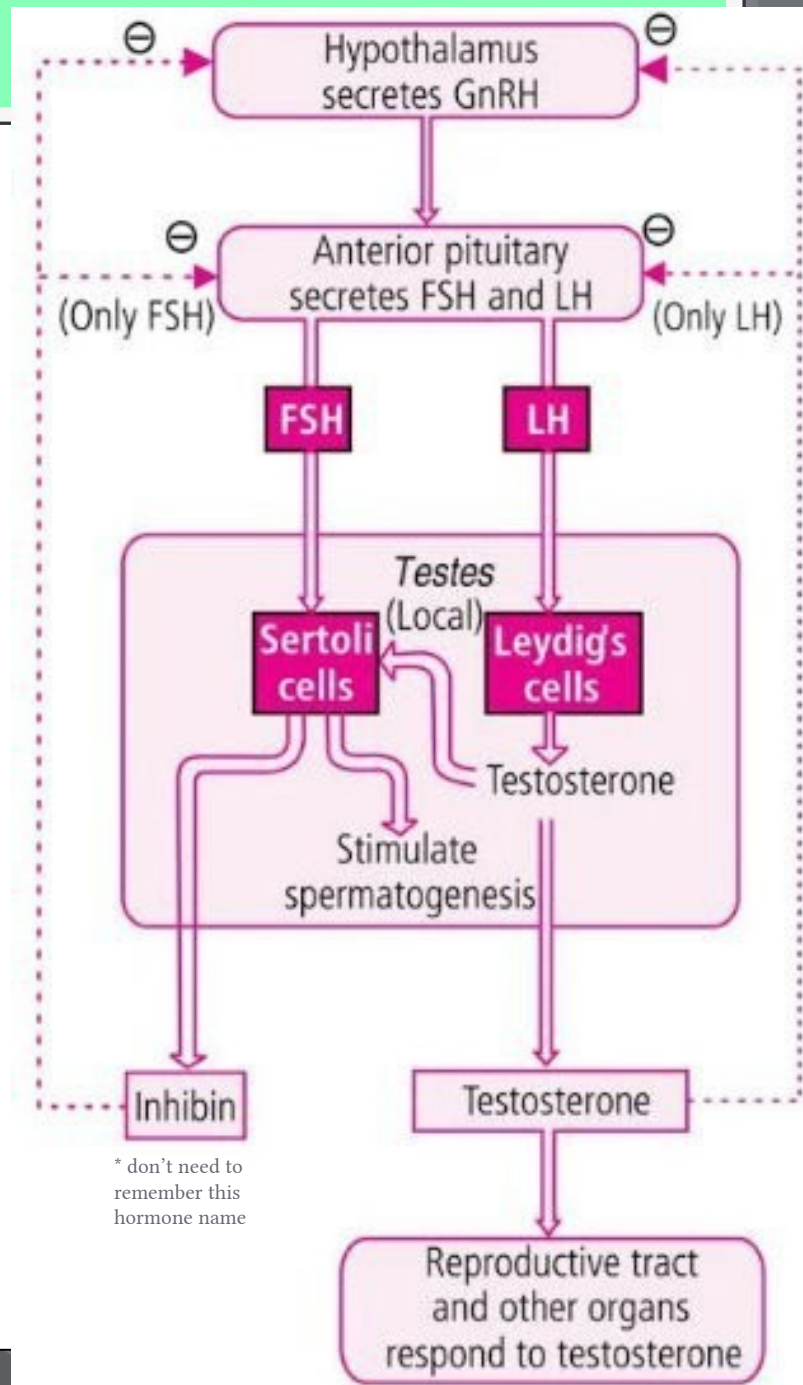
Puberty and Hormonal Regulation of Spermatogenesis

- At puberty, the hypothalamus produces **Gonadotropin Releasing Hormone (GnRH)**, which stimulates the anterior pituitary to release two hormones: **Follicle Stimulating Hormone (FSH)** and **Luteinizing Hormone (LH)**
- FSH stimulates Sertoli cells within the **seminiferous tubules** of the testes to start producing sperm
- LH stimulates the **interstitial cells** within the testes to start producing testosterone



Puberty and Hormonal Regulation of Spermatogenesis

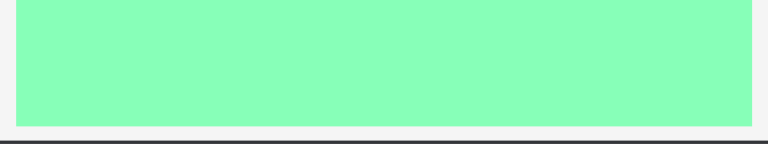
- Testosterone contributes to sperm production and also causes secondary sex characteristics to begin to develop: deepening of voice, facial hair, etc.
- Hormone levels are regulated through negative feedback



Although testosterone is especially important during puberty, it is required throughout a man's lifespan to maintain reproductive functioning and secondary sex characteristics...

- Steroids, such as hGH, can disrupt reproductive hormone feedback systems
- Side effects in men include shrinking testicles, low sperm count, loss of body hair, and the development of breasts



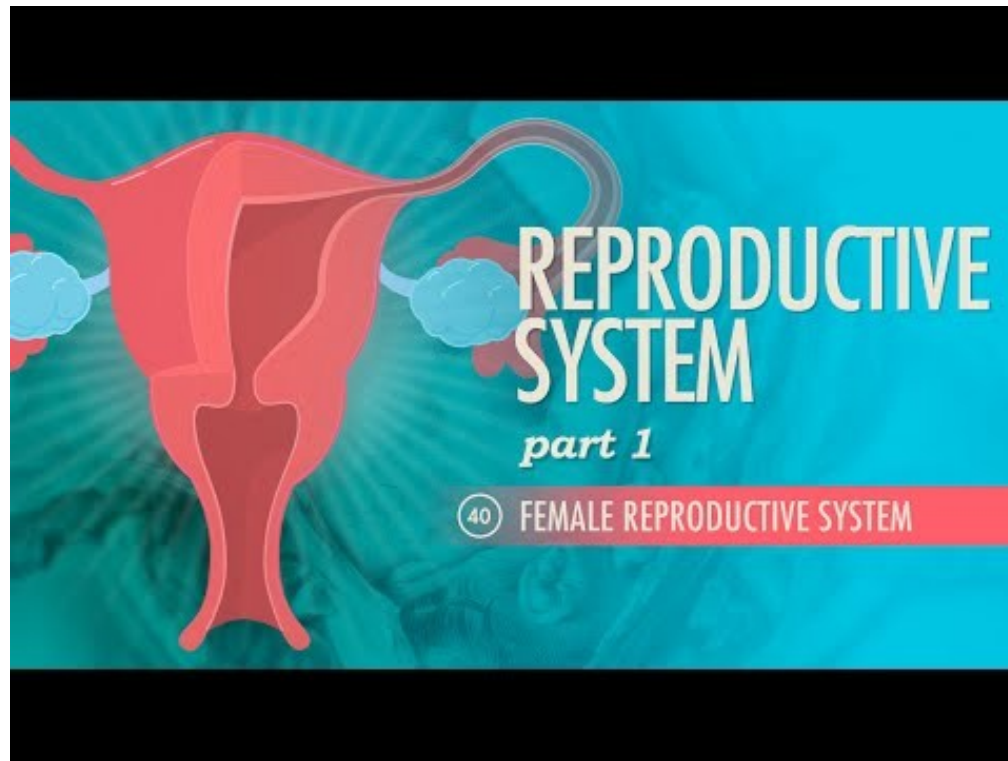


Over time, many men experience a gradual decline in testosterone levels. This condition is known as *andropause*.

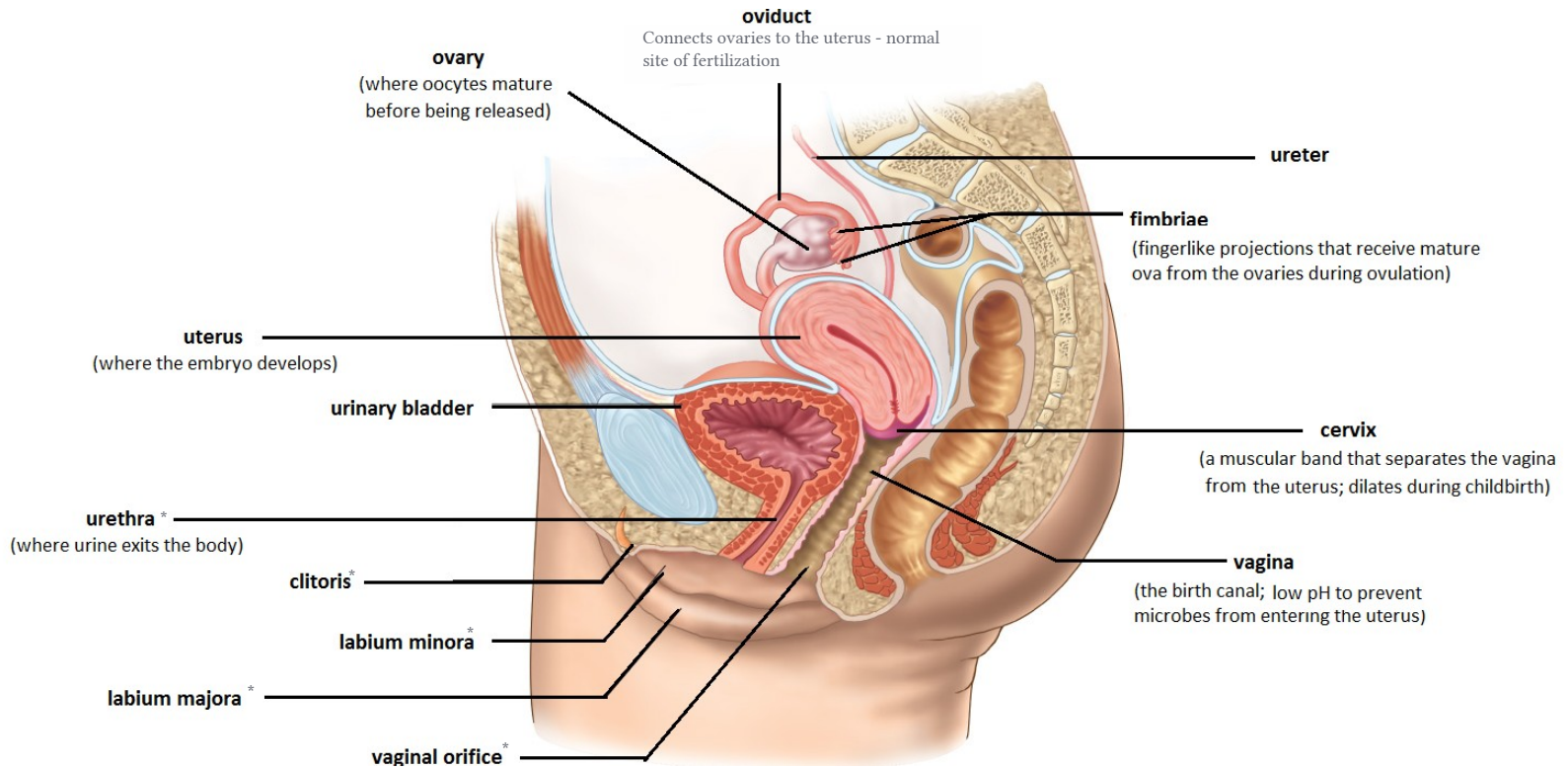
SYMPTOMS INCLUDE:

- 0 Fatigue
- 0 Depression
- 0 Loss of muscle and bone mass
- 0 Decreased sperm production

The Female Reproductive System

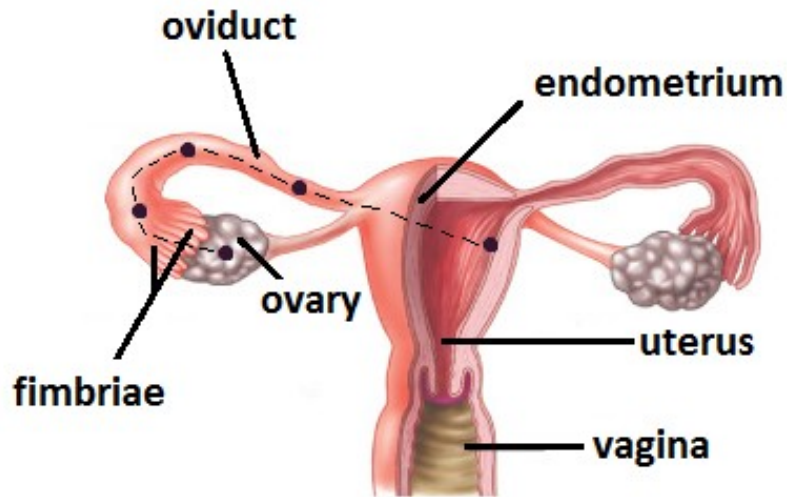


The Female Reproductive System

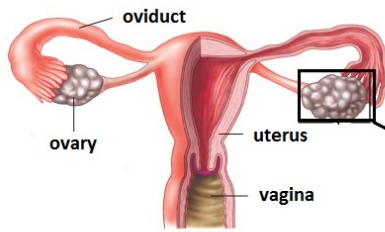


*will not be tested

The Ovaries



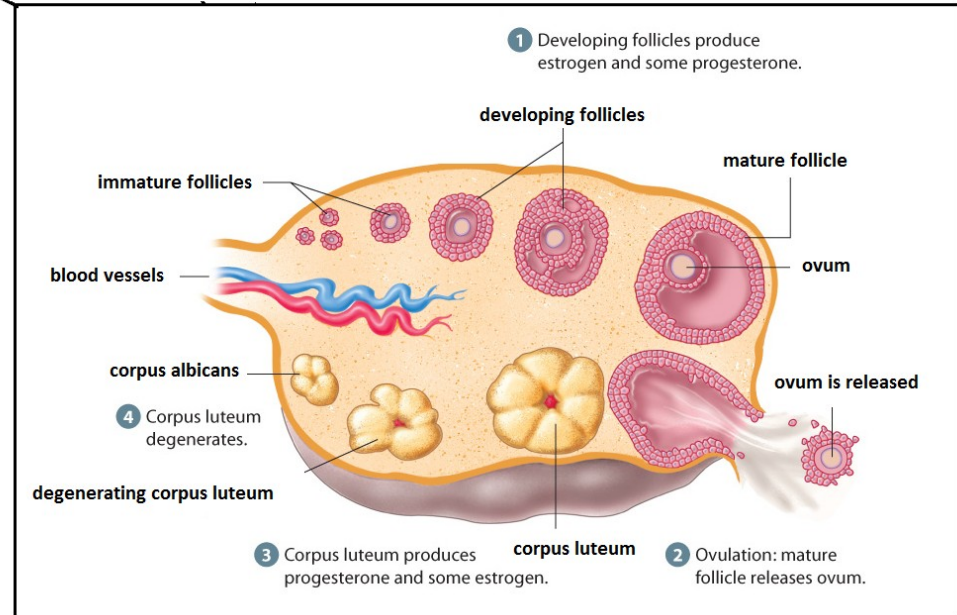
- Each month, a single mature egg (or “**ovum**”) is released from the ovary, allowing for potential fertilization. This process is known as **ovulation**.
- During this period, the **endometrial lining** of the uterus thickens in preparation for pregnancy.
- The ovum can survive in the **oviduct** for a period of 24 hours; if fertilization does not occur, the endometrial lining will then be shed. This process is known as **menstruation**.



OÖGENESIS

0 Occurs within the *follicle cells* of the ovaries; follicle cells provide nutrients necessary for the ovum's development

- The maturing follicle releases increasing amounts of **estrogen** which helps **build the endometrial lining**

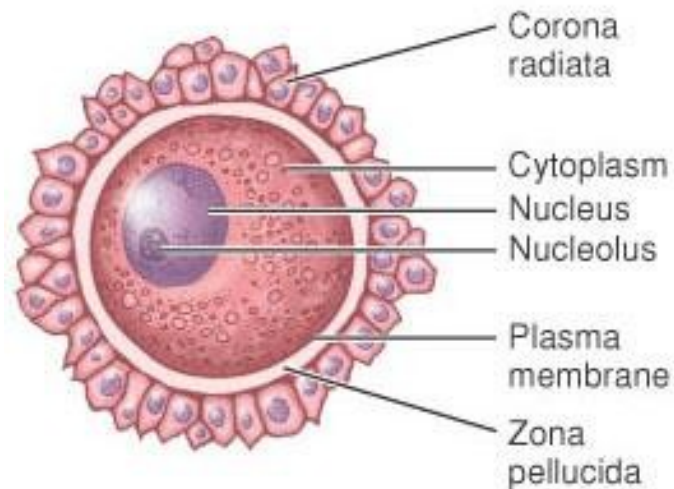


0 Once the ovum is released, surrounding follicle cells form the *corpus luteum*

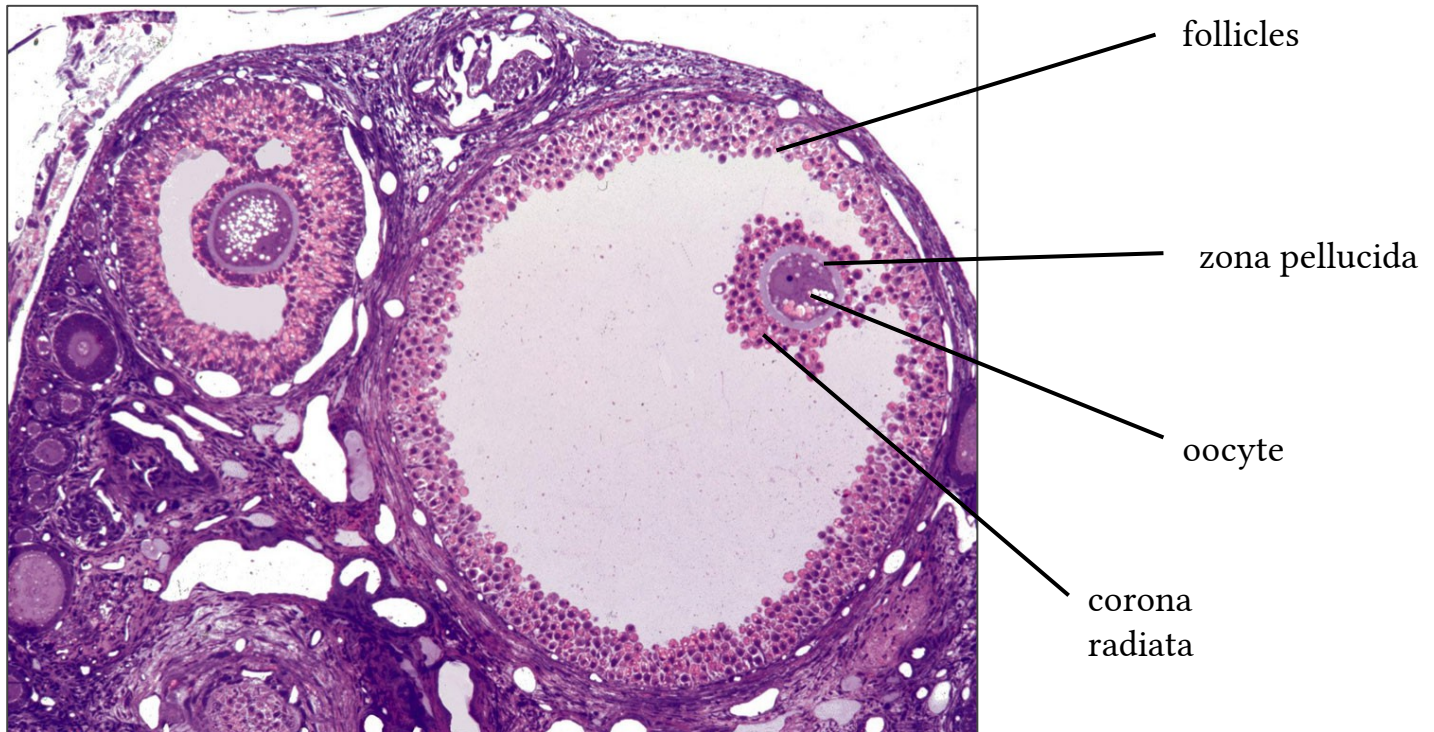
- The corpus luteum releases **some estrogen**, but **mostly progesterone** which helps **maintain the endometrial lining** in case fertilization occurs and implantation expected
- If no fertilization occurs, the corpus luteum will degenerate in about 14 days

- Females are born with a finite amount of ***oocytes***; after a certain age, then, women no can longer ovulate (menopause)
- Surrounded by two key layers (do not need to know)
 1. the ***corona radiata***, which supplies vital proteins to the cell
 2. a specialized coating called the ***zona pellucida***, which can only be penetrated by a single sperm
- Non-motile

A mature ovum:



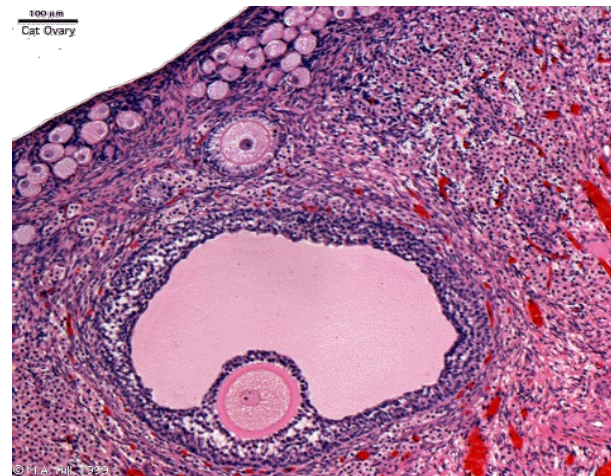
An immature oocyte developing within a follicle:



Female Sex Hormones

Like in males, puberty begins when the hypothalamus triggers the release of LH and FSH from the pituitary. However, in females...

- **FSH** stimulates the development of female sex organs as well as follicle development within the ovaries. As the follicle develops, it releases **estrogen**.
- **LH** triggers ovulation (signals egg maturation)



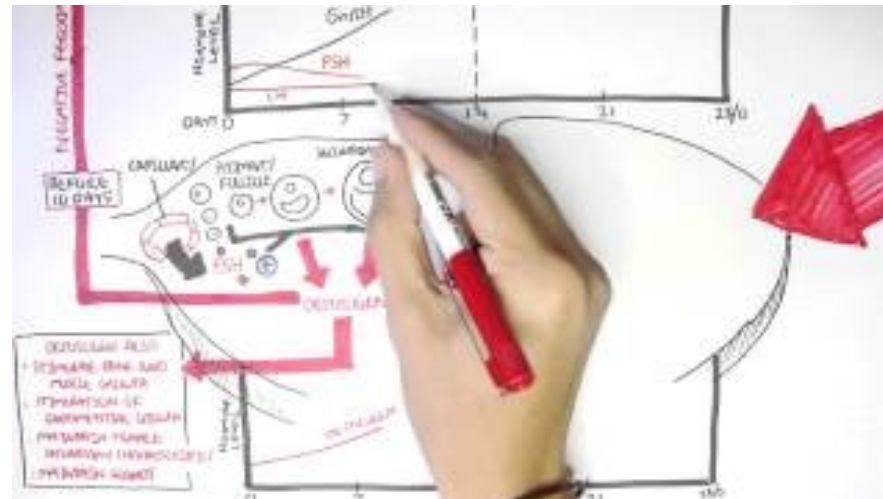
This marks the beginning of a young woman's reproductive years; associated hormones also stimulate the development of secondary sex characteristics.

Hormonal regulation of the female menstrual cycle is complex. Watch both videos to get an increasing in-depth understanding of it. There are summary notes on the next pages and a diagram, but that will not give you the whole picture.

Watch
first



Watch
second

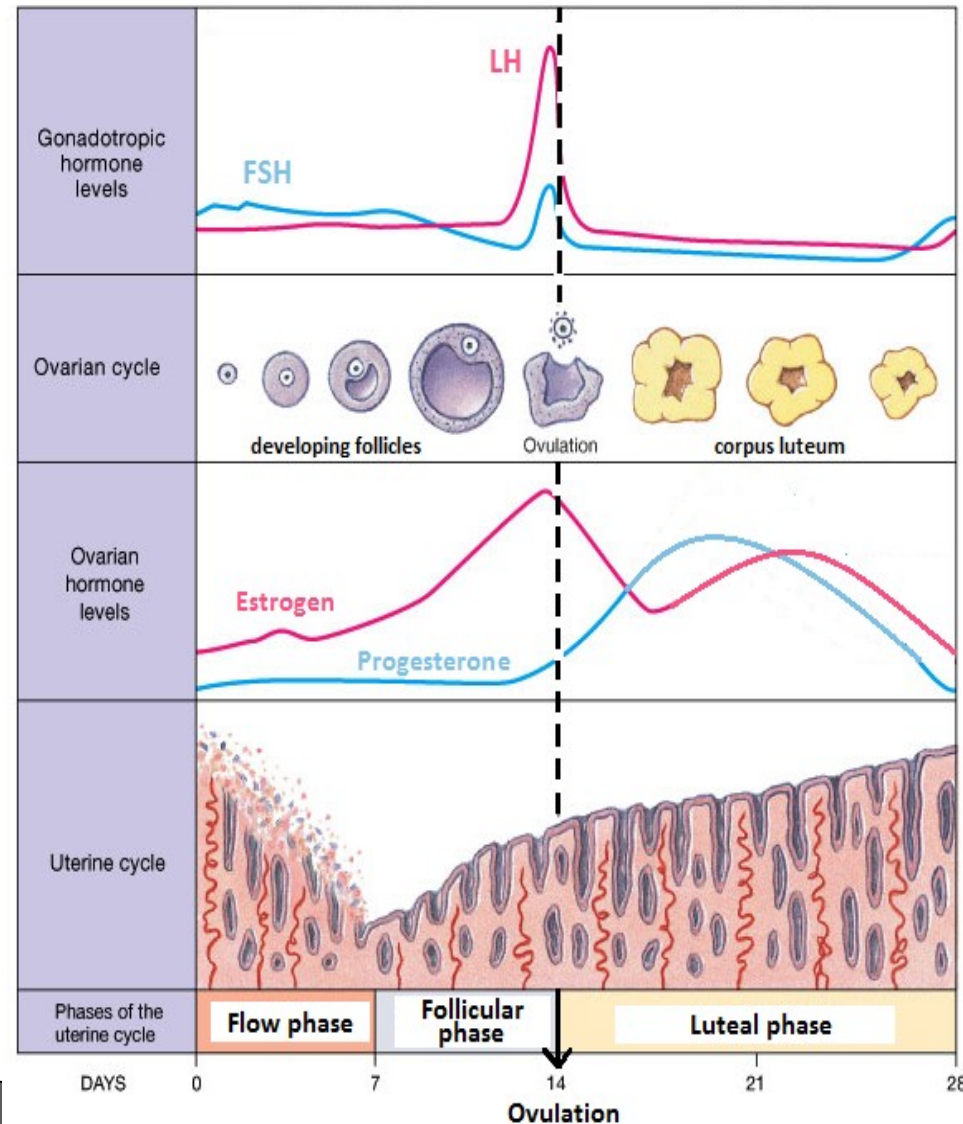


THE MENSTRUAL CYCLE

Divided into four phases...

1. Flow phase (days 0-7)

- A.k.a. Menstruation
- Marks the beginning of the cycle
- Characterized by shedding of the endometrium
- Estrogen/progesterone levels are at their lowest

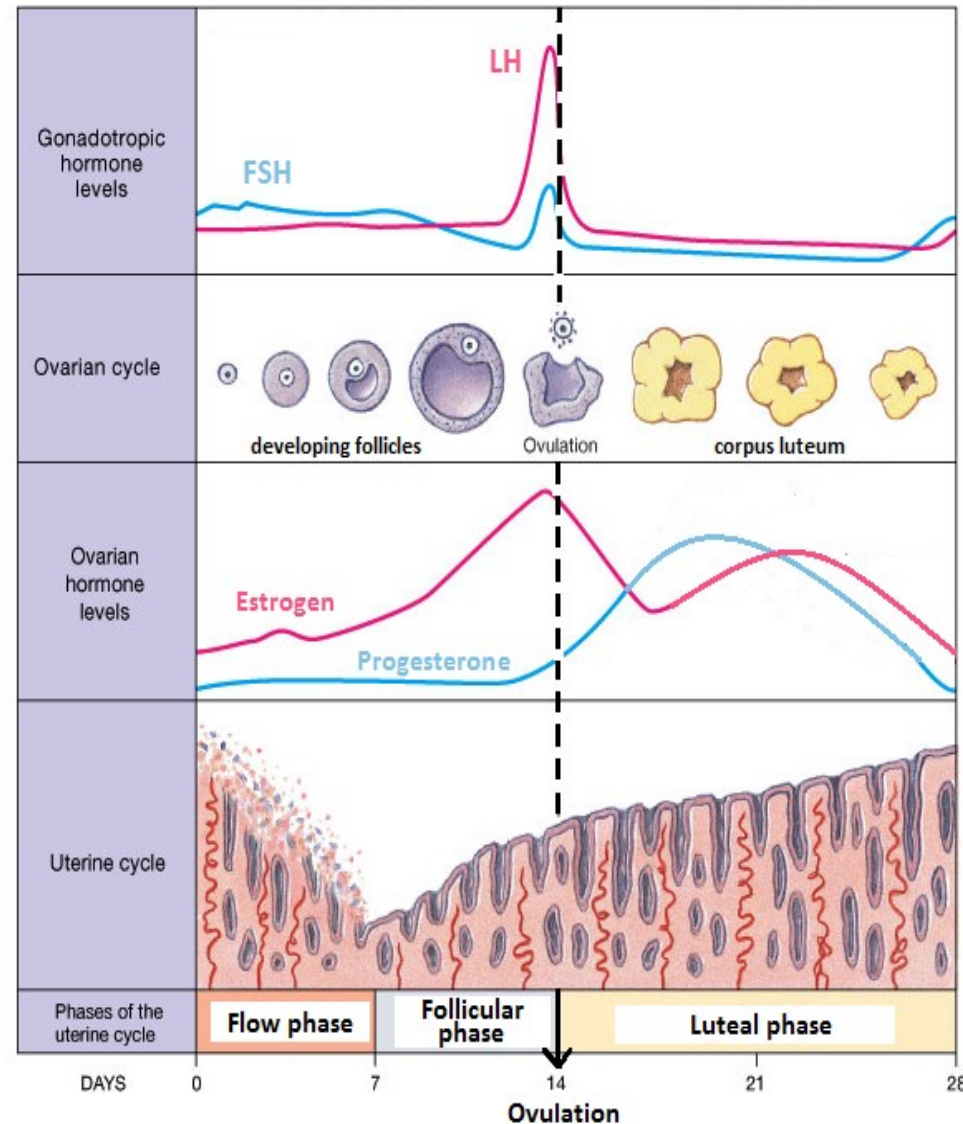


THE MENSTRUAL CYCLE

Divided into

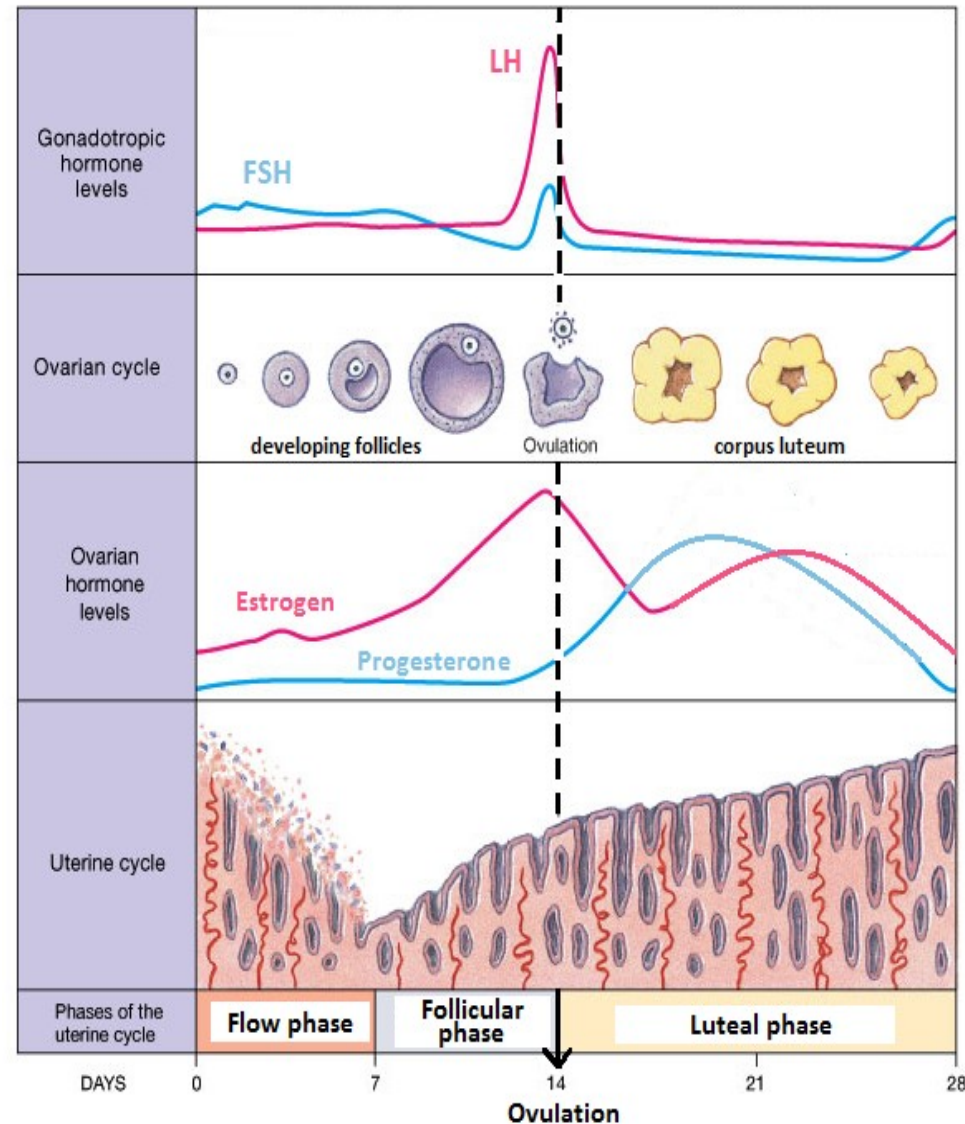
2. Follicular phase (days 7-14)

- GnRH stimulates the release of FSH from the pituitary, which causes a follicle cell within the ovary to mature.
- As the follicle develops, estrogen is secreted, promoting the development of the endometrium
- Rising levels of estrogen then inhibit the release of FSH from the pituitary



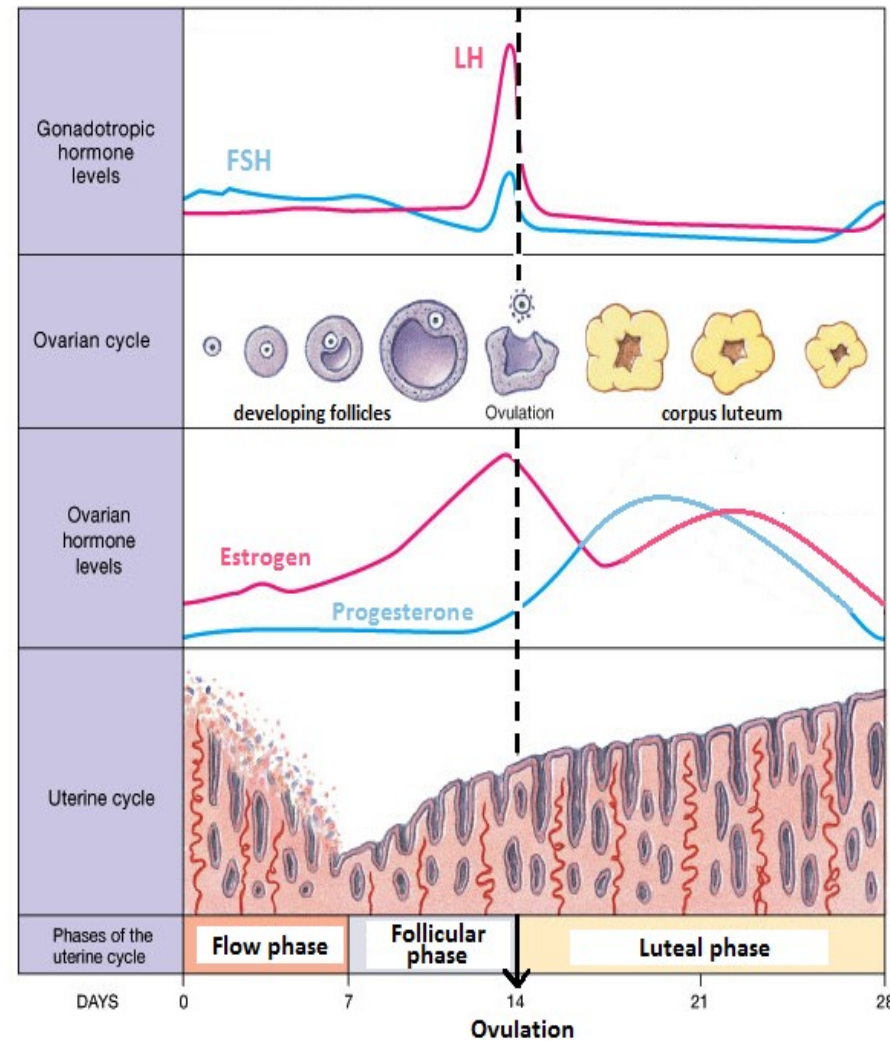
3. Ovulatory phase (day 14)

- Estrogen also triggers a sharp increase in LH production, which signals the release of the mature ovum from the follicle
- Once this occurs the follicle begins to form the corpus luteum
- Since there no no more follicle to release estrogen, we see a temporary decrease in estrogen levels



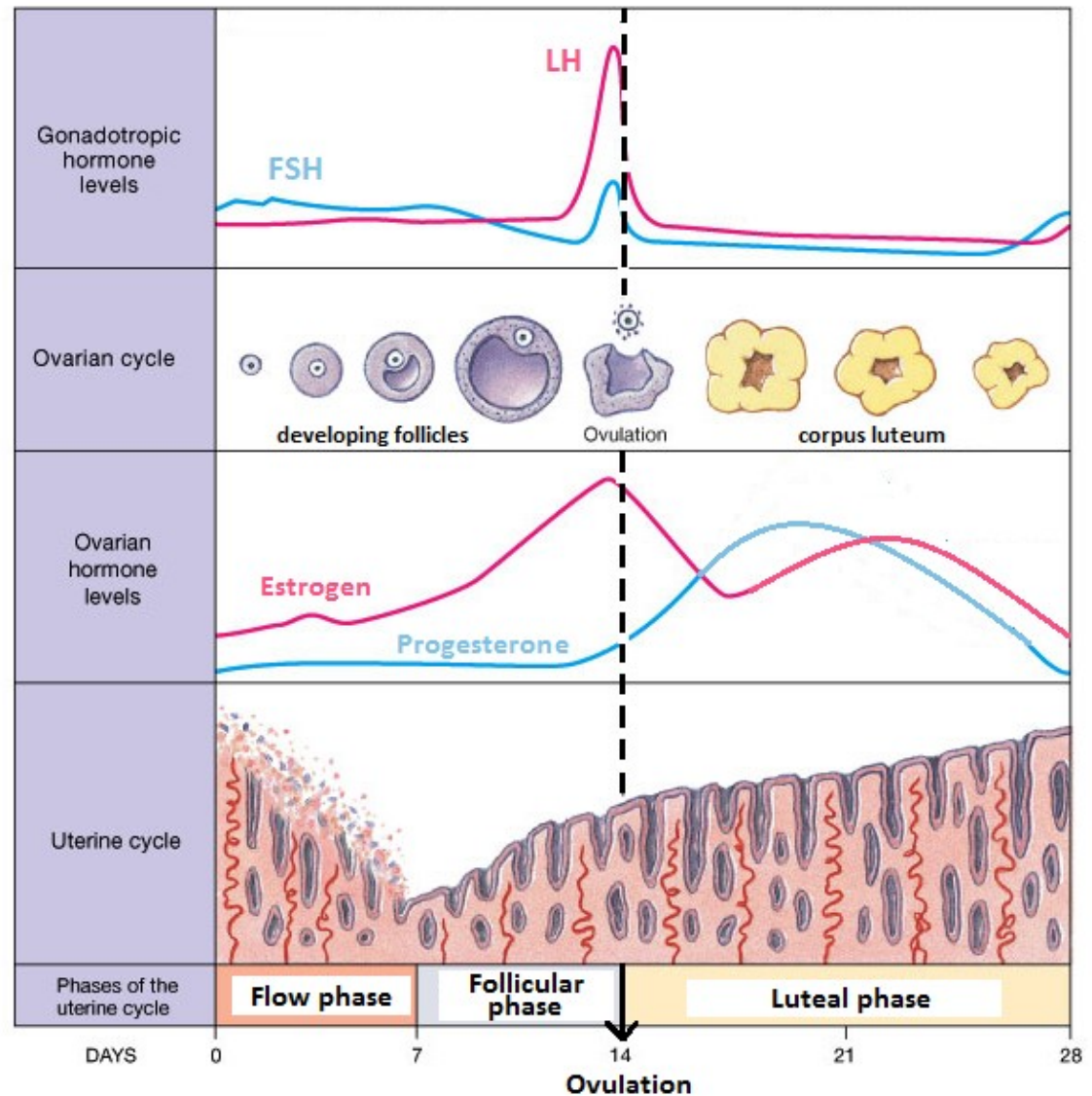
4. Luteal phase (days 14-28)

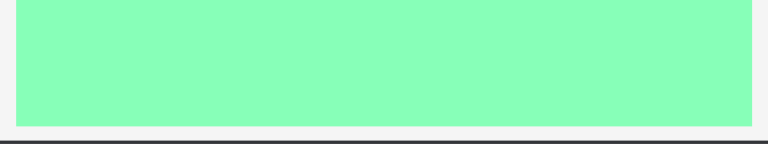
- Once the corpus luteum forms it begins to secrete estrogen and progesterone. Estrogen continues to promote the development of the endometrial lining.
- Progesterone plays an important role in preparing the uterus for an embryo. Progesterone helps maintain the endometrial lining. It also inhibits LH, preventing further ovulation and firming the cervix.
- If fertilization does not occur, the corpus luteum degenerates, and hormone secretion stops, causing the endometrial lining to be shed (beginning of flow phase)



Note: some sources only consider two separate phases, where 0-14 is follicular phase and 14-28 is luteal phase)

Pause here and try to describe all of the changes that occur during the menstrual cycle, and all causes and effects.





Eventually, women stop producing eggs. Hormone levels decline, leading to a condition known as *menopause*.

SYMPTOMS:

- “hot flashes”
- High cholesterol
- Diminished bone mass
- Changes in mood



How do birth control pills prevent pregnancy?

- Birth control pills typically contain **high doses of progesterone and estrogen**, which **inhibit release of FSH and LH through negative feedback**. Without FSH and LH, a follicle is unlikely to mature and **ovulation will not occur**.

